1 **Design & Implementation of Human-Computer Interfaces**
NPTEL-MOOCS

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2 **L31 – Empirical Study I**

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Recap

4 **Learning So Far …**

- Idea of interactive computing systems
- How to design
- How to implement

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6 **Our Next Topic**

- Empirical evaluation of system

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Empirical Study: Why

8 **Usability - Recap**

- We require an explicit measure to judge our design – if we have taken care of the human factors

The measure is called "Usability"

9 **Usability - Recap**

- ISO definition (ISO 9241-210:2009) of usability - "the extent to which a product can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use."

10 **Basic Idea**

- Earlier, we learned how to evaluate usability of our designs
  - Done by experts, mostly (on prototypes)
  - In a limited scale

11 **Basic Idea**

- We also need to evaluate our system with users, to get idea on usability

12 **Basic Idea**

- We require to collect behavioral data from users for evaluation

13 **Basic Idea**

- When we perform a controlled experiment to collect and analyze data on user behavior, the entire process is known as *empirical research*

14 **Basic Idea**

- Term not specific to study of human behavior only - any study involving observation-based data collection and analysis, for any purpose, is empirical research
- Data we get from observation is empirical data

15 **Basic Idea**

- Empirical research not easy –to obtain *reliable* empirical data that lead us to *reliable* conclusion, we have to follow a systematic process (consisting of series of stages in sequence)

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Steps for Empirical Research

17 **Stages**

- Broadly, four stages
  - Identification of research question(s)
  - Determination of variables
  - Design of experiment
  - Analysis of empirical data

18 **Understanding the Stages**

- In empirical research, we seek answer to one or more questions
  - E.g., "how good my system is" - not necessarily a *good* question
  - Finding a good question difficult and requires expertise
- We should always start by clearly specifying one or more good research question(s)

19 **Understanding the Stages**

- Next, we need to identify *variables* - to remove ambiguity in observation
  - E.g., unless we specify variables, we do not know what to observe
- We need variables to record observations for later analysis

20 **Understanding the Stages**

- Third stage: experiment (study) design - refers to planning and execution for the study

21 **Understanding the Stages**

- Getting right users and in right numbers essential - otherwise, results may not be *reliable*
- It is also necessary to determine appropriate tasks in a suitably controlled environment - without that, we may again end up with unreliable data

22 **Understanding the Stages**

- Finally, we should employ appropriate data analysis techniques to extract conclusions

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Research Question

24 **Illustrative Case Study**

- Suppose we have developed a new text entry interface (a GUI to input text)

25 **Illustrative Case Study**

- We wish to collect empirical data to measure its usability
- First step – research question

26 **Research Question**

- Consider the research question
  RQ1: *Is the new technique good?*

27 **Research Question**

- With RQ1, we are supposed to observe "if the interface we designed is good"

28 **Research Question**

- To make the observation, we set up an experiment

29 **Research Question**

- We present the interface to a user and ask him/her to judge its "quality"
- We repeat the process few times (say for five users) and complete our experiment

30 **Research Question (Recorded Observation)**

31 **Research Question**

- What can we do with the observation – not much

32 **Research Question**

- Opinion is equally divided (2 good, 2 poor/very bad, 1 in between)
- We are not likely to get answer to RQ1: is it good?

33 **Research Question**

- This is because there are elements of vagueness in both the question (RQ1) and the observations

34 **Research Question**

- Idea of interface quality (good) is vague
  - Not clear what it means (look & feel, function, both)
- User opinions can be vague (e.g., opinion of user #3)

35 **Research Question**

- We are dealing with observations that are difficult to interpret due to the vagueness inherent in the research question

36 **Note**

- We can (have to) do better!
- Shall continue in the next lecture

37 **Book**

- Bhattacharya, S. (2019). Human-Computer Interaction: User-Centric Computing for Design, McGraw-Hill India
  - Print Edition: ISBN-13: 978-93-5316-804-9; ISBN-10: 93-5316-804-X
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